M‡elYv, cix¶Y I Kbmvj‡Ubwm †K›`
ci I cwi‡ek ‡KŠkj wefvM
kvnRvjvj weÁvb I chw³ wekwe`¨vjq
wm‡jU-3114, evsjv‡`k
wcGweG · :0821-713491/714479/713850
(G · ‡Ubkb 269)

Center for Research, Testing and Consultancy
Department of Civil and Environmental Engineering
Shahjalal University of Science and Technology
Sylhet–3114, Bangladesh
PABX :0821-713491/714479/713850 (Extension 269)
Fax :880-821-715257, Email: cee@sust.edu

Ref. no: CRTC/CEE/ G20221038/ S- 176

SUB-SOIL INVESTIGATION REPORT

PROPOSED FOR 05 STORIED RESIDENTIAL BUILDING
AT TEMUKHI, KUMARGAON, SYLHET
DAG # 374, KHA # 9536, J.L # 80, MOUJA- KUMARGOAN

CLIENT

JOSIM UDDIN
S/O- JOYNAL ABEDIN

Verification Code: 9SXc9b42
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET.

SAMPLE COLLECTED BY

M/S TECHNO SOIL INVESTIGATIOR
Mokothchhor Market, Bagbari Barnomala Point, Sylhet.
Mobile: 01712-710546, 01612710546

Introduction

A reasonably accurate conception of the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study of the sub-surface elements in this regard would be necessary to select the proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing, and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Borehole no	Depth	Quantities
01	Disturbed sample	01	0-75 ft	15
02	Disturbed sample	02	0-70 ft	14
03	Disturbed sample	03	0-65 ft	13
04	Disturbed sample	04	0-55 ft	11
05	-	-	-	-
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with borehole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, the Bearing Capacity of soil samples, including skin friction & end bearing, was calculated.

Bearing capacity for shallow foundation

The N value gives valuable information regarding the consistency of cohesive soil and relative density of cohesion less soil. These relationships are shown in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less soil	
$q_{ult} = 30N;$ $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

Table 7: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	3	0.312	0.328	0.358	0.368	0.398	0.433	0.336
BH02	2.5-5.0	3	0.394	0.415	0.452	0.466	0.503	0.547	0.425
	5.0-10	4	0.512	0.538	0.587	0.605	0.653	0.711	0.552
BH03	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	5	0.588	0.618	0.674	0.694	0.750	0.816	0.634
BH04	2.5-5.0	3	0.325	0.342	0.372	0.384	0.414	0.451	0.350
	5.0-10	4	0.508	0.534	0.582	0.600	0.648	0.705	0.547
BH05	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH06	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH07	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH08	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH09	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-
BH10	2.5-5.0	-	-	-	-	-	-	-	-
	5.0-10	-	-	-	-	-	-	-	-

Table 8: The allowable skin friction for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.074	0.049	0.049	-	-	-	-	-	-
10	0.049	0.049	0.074	0.074	-	-	-	-	-	-
15	0.049	0.118	0.049	0.074	-	-	-	-	-	-
20	0.096	0.138	0.218	0.156	-	-	-	-	-	-
25	0.156	0.156	0.239	0.120	-	-	-	-	-	-
30	0.096	0.156	0.308	0.144	-	-	-	-	-	-
35	0.118	0.118	0.174	0.112	-	-	-	-	-	-
40	0.156	0.104	0.196	0.152	-	-	-	-	-	-
45	0.118	0.138	0.112	0.168	-	-	-	-	-	-
50	0.118	0.156	0.128	0.200	-	-	-	-	-	-
55	0.174	0.230	0.136	0.192	-	-	-	-	-	-
60	0.218	0.254	0.184	-	-	-	-	-	-	-
65	0.156	0.184	0.184	-	-	-	-	-	-	-
70	0.176	0.176	-	-	-	-	-	-	-	-
75	0.168	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	0.675	0.450	0.450	-	-	-	-	-	-
10	0.450	0.450	0.675	0.675	-	-	-	-	-	-
15	0.450	1.125	0.450	0.675	-	-	-	-	-	-
20	0.900	1.350	2.250	1.575	-	-	-	-	-	-
25	1.575	1.575	2.475	24.000	-	-	-	-	-	-
30	0.900	1.575	5.175	28.800	-	-	-	-	-	-
35	1.125	1.125	1.800	22.400	-	-	-	-	-	-
40	1.575	20.800	2.025	30.400	-	-	-	-	-	-
45	1.125	1.350	22.400	33.600	-	-	-	-	-	-
50	1.125	1.575	25.600	40.000	-	-	-	-	-	-
55	1.800	3.150	27.200	38.400	-	-	-	-	-	-
60	2.250	4.275	36.800	-	-	-	-	-	-	-
65	1.575	36.800	36.800	-	-	-	-	-	-	-
70	35.200	35.200	-	-	-	-	-	-	-	-
75	33.600	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.048	0.032	0.032	-	-	-	-	-	-
10	0.032	0.032	0.048	0.048	-	-	-	-	-	-
15	0.032	0.077	0.032	0.048	-	-	-	-	-	-
20	0.062	0.090	0.141	0.101	-	-	-	-	-	-
25	0.101	0.101	0.156	0.078	-	-	-	-	-	-
30	0.062	0.101	0.200	0.094	-	-	-	-	-	-
35	0.077	0.077	0.113	0.073	-	-	-	-	-	-
40	0.101	0.068	0.127	0.099	-	-	-	-	-	-
45	0.077	0.090	0.073	0.109	-	-	-	-	-	-
50	0.077	0.101	0.083	0.130	-	-	-	-	-	-
55	0.113	0.149	0.088	0.125	-	-	-	-	-	-
60	0.141	0.165	0.120	-	-	-	-	-	-	-
65	0.101	0.120	0.120	-	-	-	-	-	-	-
70	0.114	0.114	-	-	-	-	-	-	-	-
75	0.109	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.304	0.203	0.203	-	-	-	-	-	-
10	0.203	0.203	0.304	0.304	-	-	-	-	-	-
15	0.203	0.506	0.203	0.304	-	-	-	-	-	-
20	0.405	0.608	1.013	0.709	-	-	-	-	-	-
25	0.709	0.709	1.114	10.800	-	-	-	-	-	-
30	0.405	0.709	2.329	12.960	-	-	-	-	-	-
35	0.506	0.506	0.810	10.080	-	-	-	-	-	-
40	0.709	9.360	0.911	13.680	-	-	-	-	-	-
45	0.506	0.608	10.080	15.120	-	-	-	-	-	-
50	0.506	0.709	11.520	18.000	-	-	-	-	-	-
55	0.810	1.418	12.240	17.280	-	-	-	-	-	-
60	1.013	1.924	16.560	-	-	-	-	-	-	-
65	0.709	16.560	16.560	-	-	-	-	-	-	-
70	15.840	15.840	-	-	-	-	-	-	-	-
75	15.120	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

Based on analysis and information provided by the owner, the following recommendations are suggested for the proposed five (05) storied residential buildings.

Soil profile scenario

The overall soil information of the investigated site is quite irregular among the boreholes, and the soil formation is vertically inconsistent. There have been nearly two distinct types of drilling scenarios observed.

For type A (borehole no.: 01, & 04)- gray soft clay with low bearing capacity exists up to 50 ft and 20 ft from the EGL in boreholes no.:01 and 04, respectively. A layer of gray silty clay with good bearing capacity was found from 55-65 ft depth of EGL in borehole no.:01. Finally, gray fine sand with silt having good bearing capacity begins at 70 ft and 25 ft of EGL in boreholes no.:01 and 04, respectively and continues after that.

For type B (borehole no.: 02 & 03)- gray soft clay with low bearing capacity exists up to 35 ft and 15 ft from the EGL in boreholes no.:02 and 03, respectively. After that, gray fine sand with silt and gray silty clay having moderate bearing capacity have been observed. Although gray soft clay with low bearing capacity has been found, the layer did not go down to the last. The gray fine sand with silt clay layer with good bearing capacity begins at 65 ft and 45 ft of EGL, respectively, in boreholes no. 02 and 03, and extends to a greater depth in the borehole.

From the above discussion, it is apparent that the proposed site's whole scenario is quite different.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. Shallow foundations may not be suitable for a five-story residential building because most boreholes contain soft clay soil with low bearing capacity at the top. Thus, the soft clay soil layer is highly susceptible to settlement.#The bearing capacity for the shallow foundation given in **table 7** is based on existing soil conditions.

However, deep foundation may be preferentially recommended to ensure the stability of the structure. The size and depth of the pile should be selected in light of the bearing capacity given in the report.

Safety provisions

It is strongly recommended to check the **site's settlement** considering the building load while designing either a shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out to confirm the service pile's design parameters (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work, including workmanship, should be considered for the safe design of a shallow or deep foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values are shown in the Appendix. The client performed no particular laboratory test for the proposed site.

Special notes

These suggestions are absolutely tentative, and the selection of type, depth, and dimensions of the foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **75** ft, as the client provided.

Report Prepared By	**Test Assisted By**	**Counter Signed By**
Md. Aminul Islam	**A H M Humayun Kabir**	**Prof Dr. H M A Mahzuz**
Assistant Professor Dept. of CEE, SUST	Senior Technical Officer Dept. of CEE, SUST	Coordinator, CRTC Dept. of CEE, SUST